

1. Create a global variable `total` set to 0. Write a function `add_to_total` that uses the `global` keyword to add a given number to `total`. Call it with 10 and then 20, and print the final `total`.

2. Create a global variable `price` set to 500. Write a function `apply_discount` that takes a `price` parameter, subtracts 50 from it, and prints the result. Call it with `price`, then print `price` again afterward and explain in a comment why it is unchanged.

*Hint: the parameter `price` inside the function is a separate local variable from the global one.*

3. Write a function `show_counter` that creates a local variable `counter` set to 1 and prints it. Call the function twice in a row and explain in a comment why it prints 1 both times, not 1 then 2.

*Hint: local variables do not remember anything between separate calls.*

4. Create a global variable `balance` set to 1000. Write a function `withdraw` that takes an `amount` parameter and tries to do `balance = balance - amount`, without using the `global` keyword. Run it, note the exact error Python shows, then fix it.

*Hint: assigning to `balance` inside the function makes Python treat it as local for the whole function.*

5. Write two separate functions that each create a local variable named `result` but give it a different value and print it. Call both functions one after another and confirm they never interfere with each other.